HDOC DAY-65

Write a recursive program to solve the **Tower of Hanoi** problem for **n** disks and display all moves.

Algorithm:

1. Start.
2. Input the number of disks n.
3. Consider:
    - A = Source rod
    - B = Auxiliary rod
    - C = Destination rod
4. If n == 1, move the disk directly from source to destination.
5. Otherwise:
    - Move n-1 disks from **source to auxiliary** using destination.
    - Move disk n from **source to destination**.
    - Move n-1 disks from **auxiliary to destination** using source.
6. Repeat these steps recursively until the base case n == 1 is reached.
7. Stop.

- ## C Program:

```
#include <stdio.h>

void towerOfHanoi(int n, char source, char auxiliary, char destination)
{
    if (n == 1)
    {
        printf("Move disk 1 from %c to %c\n", source, destination);
        return;
    }

    // Move n-1 disks from source to auxiliary
    towerOfHanoi(n - 1, source, destination, auxiliary);

```

```
    // Move the largest disk from source to destination
    printf("Move disk %d from %c to %c\n", n, source, destination);

    // Move n-1 disks from auxiliary to destination
    towerOfHanoi(n - 1, auxiliary, source, destination);
}

int main()
{
    int n;

    printf("Enter the number of disks: ");
    scanf("%d", &n);

    towerOfHanoi(n, 'A', 'B', 'C');

    return 0;
}
```

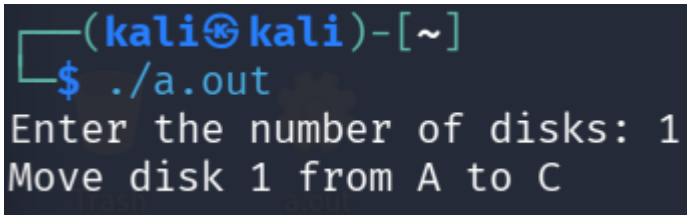
Test Case Table:

Test Case	Input n	Expected Number of Moves	Result
1	1	1	Pass
2	2	3	Pass
3	3	7	Pass
4	4	15	Pass
5	5	31	Pass

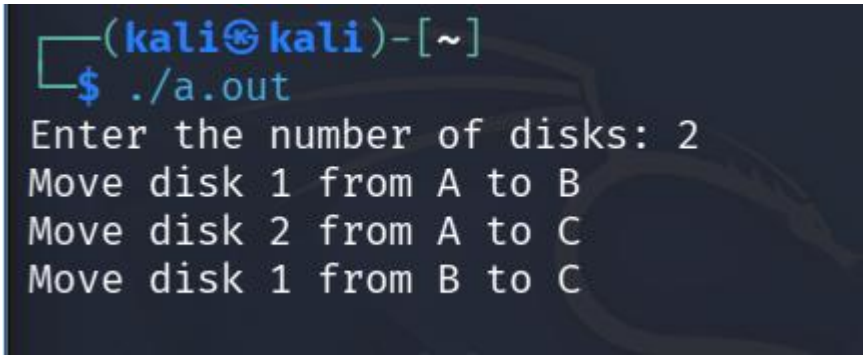
Compilation and Execution:

The following program has been compiled and executed successfully.

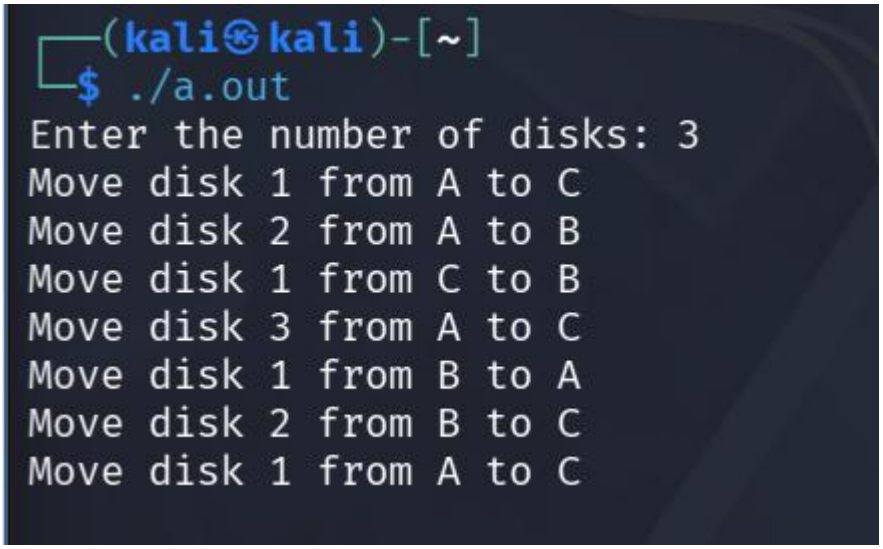
Test Case 1:



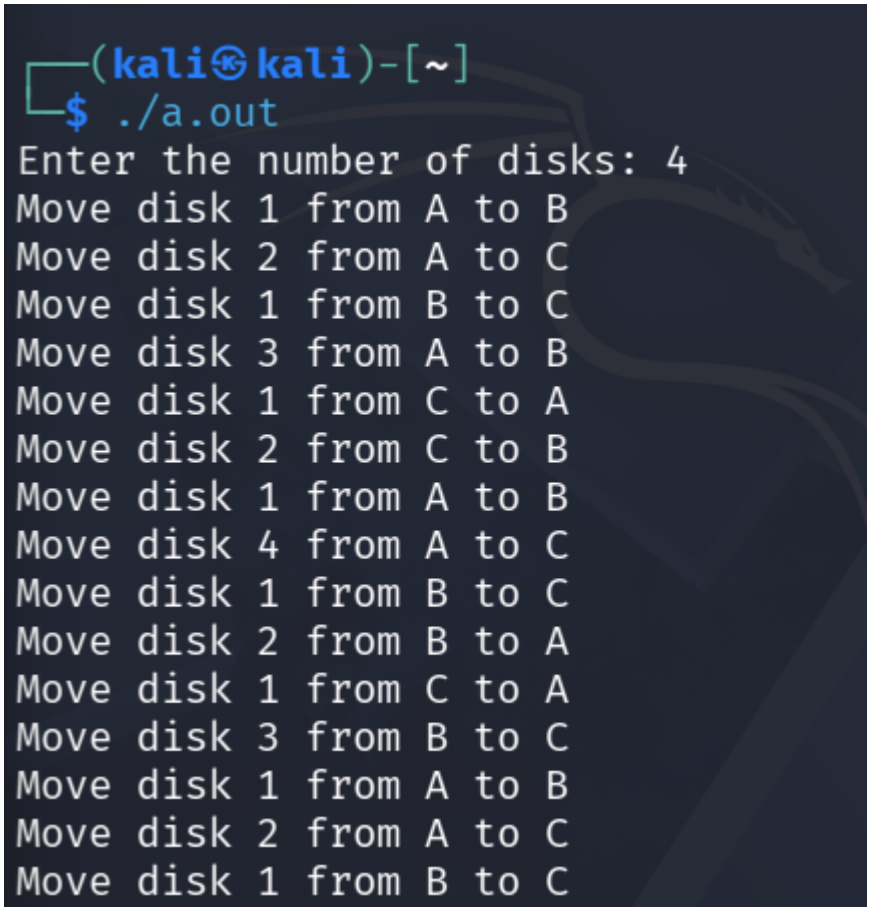
Test Case 2:



Test Case 3:



Test Case 4:


(kali㉿kali)-[~]
$ ./a.out
Enter the number of disks: 4
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from B to C
Move disk 3 from A to B
Move disk 1 from C to A
Move disk 2 from C to B
Move disk 1 from A to B
Move disk 4 from A to C
Move disk 1 from B to C
Move disk 2 from B to A
Move disk 1 from C to A
Move disk 3 from B to C
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from B to C

Test Case 5:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter the number of disks: 5
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
Move disk 4 from A to B
Move disk 1 from C to B
Move disk 2 from C to A
Move disk 1 from B to A
Move disk 3 from C to B
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 5 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
Move disk 3 from B to A
Move disk 1 from C to B
Move disk 2 from C to A
Move disk 1 from B to A
Move disk 4 from B to C
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
```